



Profile

Electrical Engineering graduate focused on embedded software, microcontroller-based development, test automation, and AI-assisted validation. Experienced in Python, Robot Framework, embedded Linux, CI/CD, and intelligent diagnostic solutions for reliable software systems.

Experience

STMicroelectronics – Final Year Internship

Feb. 2026 – Aug. 2026

- Designed a modular Python and Robot Framework solution for automated STM32 secure-boot validation.
- Automated build, provisioning, hardware reset, UART capture, full regression, and test reporting workflows.
- Reduced validation time by up to 48% (approximately 15 to 8 minutes), enabling continuous, repeatable testing.
- Built AI-assisted failure diagnosis from Robot Framework, UART, and subprocess logs using structured JSON incidents, local diagnostic rules, and LLM API integration.

FABLAB ENIM – Engineering Internship

Jul. 2025 – Aug. 2025

- Developed a prototype for motor energy monitoring on a microcontroller platform.
- Deployed an embedded Linux monitoring solution using Yocto on Raspberry Pi 4.
- Integrated sensors and MQTT communication for an embedded IoT supervision system.
- Technologies:** STM32, sensors, Yocto, Raspberry Pi 4, MQTT, IoT.

STEG Manzel Bouzelfa – Introductory Internship

Jul. 2024 – Aug. 2024

- Inspected industrial electrical systems while applying safety procedures.
- Supported maintenance observations in an industrial environment.
- Areas:** Industrial electrical systems, maintenance, safety.

Technical Projects

Multi-sensor IoT Embedded System based on Yocto – Real-time monitoring with data acquisition, MQTT communication, alarms, and logging.

Technologies: Yocto Linux, IoT, MQTT, Python, QEMU

Driver Assistance System – STM32-based module integrating GPS and LiDAR for navigation and obstacle detection.

Technologies: STM32, LiDAR, GPS, graphical interface

Real-time Sensor Fusion – Acquisition, filtering, and processing of IMU data with communication to embedded Linux.

Technologies: STM32, IMU, FreeRTOS, I2C, SPI, embedded Linux

STM32H7-based Electrical Control Cabinet – Control cabinet with TouchGFX touchscreen interface for supervision.

Technologies: STM32H7, TouchGFX, embedded HMI, industrial control

Technical Skills

- Programming & Scripting: C, C++, Python, Shell
- Embedded Software: STM32 (HAL/LL), ARM Cortex-M, FreeRTOS, secure boot, sensor/actuator interfacing
- Test Automation: Robot Framework, Selenium, PyAutoGUI, regression testing, log analysis, automated reporting
- AI & Machine Learning: LLM API integration, AI-assisted diagnostics, incident classification, Machine Learning
- DevOps & Version Control: Git, GitLab, GitHub, Jenkins, CI/CD, continuous testing
- Embedded Linux & IoT: Yocto, QEMU, MQTT, Linux scripting
- Protocols & Tools: UART, SPI, I2C, CAN, LoRaWAN, STM32CubeIDE, PyCharm, VS Code, Qt, Simulink

Certifications

- Learning Git and GitHub – LinkedIn Learning
- Jul. 2026
- Learning Jenkins – LinkedIn Learning
- May 2026
- Introduction to Linux (LFS101) – The Linux Foundation
- Nov. 2025
- Open Source RT-Thread RTOS on RISC-V (LFD123) – The Linux Foundation
- Nov. 2025
- Foundational C# with Microsoft – freeCodeCamp / Microsoft
- Jun. 2024

Education

ENIM – National Engineering School of Monastir

2023 – 2026

Electrical Engineering, Monastir, Tunisia

IPEIEM – Preparatory Institute for Engineering Studies of El Manar

2021 – 2023

Physics & Chemistry, Tunis, Tunisia

Leadership & Languages

ING.ENIM Robotics Club

Jun. 2024 – Jun. 2025

President

Monastir, Tunisia

Organized events and provided technical mentoring in embedded systems and robotics.

Languages: French: Fluent | English: Fluent | Arabic: Native